Chapter 26

The Two Person Bargaining Problem

The Nash bargaining theory, based on the brilliant work of John Nash, represents one of the earliest and most influential results in cooperative game theory. Given two rational and intelligent players and a set of feasible allocations from among which a unique allocation is to be chosen, the Nash bargaining theory proposes an elegant axiomatic approach to solve the problem. This chapter sets up the problem, presents the axioms of Nash bargaining theory, and provides a proof of the Nash bargaining result.

26.1 Nash Program

Cooperation refers to coalitions of two or more players acting together with a specific common purpose in mind. Since rationality and intelligence are two fundamental assumptions in game theory, any cooperation between players must take into account the objective of maximizing their own individual payoffs. As we have seen in the previous chapter, the notion of cooperation which is closely tied with the notion of correlated strategies can be developed without abandoning the basic tenets underlying non-cooperative game theory. This has been emphasized by John Nash himself [1, 2]. According to Nash, cooperative actions can be considered as the culmination of a certain process of bargaining among the cooperating players and consequently, cooperation between players can be studied using core concepts of non-cooperative game theory. In this bargaining process, we can expect each player to behave according to some bargaining strategy that satisfies the original utility maximization criterion as in standard non-cooperative game theory.

The ingenious idea of Nash is to define a *cooperative transformation* that will transform a strategic form game into another strategic form game that has an extended strategy space for each player. The extended strategy set for a player has all the strategies of the original game and also additional strategies that capture bargaining with the other players to jointly plan cooperative strategies. This is on the lines of what we have studied in the previous chapter on games with contracts and games with communication. We will illustrate this with the standard example of the prisoner's dilemma problem which provides a classic example for illustrating the benefits of cooperation.

	NC	C
NC	$-2, -2$	$-10, -1$
C	$-1, -10$	$-5, -5$

Clearly, the rationality of the players suggests the equilibrium (C, C) which is obviously not an optimal option for the players. The two players here have a strong incentive to bargain with each other or sign a contract to transform this game into one which has equilibria that are better for both the players. The concept of a *Nash program* for cooperative game theory is to define cooperative solution concepts such that a cooperative solution for any given game is a Nash equilibrium of some cooperative transformation of the original non-cooperative game. The notion of Nash program was introduced by Nash in his classic paper [2]. If we carefully describe all the possibilities that are feasible when the players bargain or sign contracts with each other, we may end up with a game that has numerous equilibria. This means the Nash program may not lead to a unique cooperative solution. This means that we should have a sound theory of cooperative equilibrium selection. This is what the Nash bargaining theory offers in an axiomatic and rigorous, but in a highly intuitive way.

26.2 The Two Person Bargaining Problem

According to Nash, the term *bargaining* refers to a situation in which

- two individuals have the possibility of concluding a mutually beneficial agreement
- there is a conflict of interest about which agreement to conclude
- no agreement may be imposed on any player without that player's approval.

The following assumptions are implicit in Nash's formulation: when two players negotiate or an impartial mediator arbitrates, the payoff allocations that the two players ultimately get should depend only on:

- the payoffs they would expect if the negotiation or arbitration were to fail to reach a settlement, and
- the set of payoff allocations that are jointly feasible for the two players in the process of negotiation or arbitration.

The two person bargaining problem has been applied in many important contexts including:

- *Management labor arbitration* where the management negotiates contracts with the labor union
- *International relations*, where two nations or two groups of nations work out agreements on issues such as nuclear disarmament, military cooperation, anti-terrorist strategy, bilateral emission control initiatives, etc.

- *Duopoly market games* where two major competing companies work out adjustments on their production to mutually maximize their profits
- *Bilateral trade situations* where a buyer and seller engage in bargaining over the trade
- *Supply chain contracts* where a buyer and a supplier work out a mutually beneficial contract that facilitates a long-term relationship
- *Negotiation protocols* in multiagent systems
- *Property settlement disputes* between corporations or individuals

The Bargaining Problem

The two person bargaining problem consists of a pair (F, v) where F is called the feasible set and v is called the disagreement point.

- F, the feasible set of allocations, is a closed, convex subset of $\mathbb{R}^2$.
- The disagreement point $v = (v_1, v_2) \in \mathbb{R}^2$ represents the disagreement payoff allocation for the two players. It is also called the *status-quo point* or the *default point* or *de facto point*. This gives the payoffs for the two players in the event that the negotiations fail. It may be noted that v is invariably chosen to belong to the feasible set F.
- The set $F \cap \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq v_1; x_2 \geq v_2\}$ is assumed to be non-empty and bounded.

Rationale Behind the Assumptions

- Convexity of F: This has the following rationale. It is reasonable for the two players to agree to correlated strategies. Consequently, if the utility allocations $x = (x_1, x_2)$ and $y = (y_1, y_2)$ are feasible and $0 \leq \lambda \leq 1$, then the expected utility allocation $\lambda x + (1 - \lambda)y$ can be achieved by planning to implement a correlated strategy which assigns probability λ to $x = (x_1, x_2)$ and probability $(1 - \lambda)$ to $y = (y_1, y_2)$.
- Closedness of F: This means any convergent sequence in F will converge to a point that belongs to F. It would be strange if we have a sequence of allocations belonging to F and the limiting allocation does not belong to F. In fact, without this assumption, it may happen that there may not exist a solution to the bargaining problem since the limit of a sequence may not be in F. Closedness of F is therefore a natural topological requirement.
- The third assumption is the set $F \cap \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq v_1, x_2 \geq v_2\}$ is non-empty and bounded. This assumption implies that (a) there exists some feasible allocation that is at least as good as the disagreement point for both players, however (b) unbounded gains over the disagreement point are not possible. Both these requirements are clearly reasonable.

Connection to Two Player Non-Cooperative Games

Suppose $\Gamma = \langle \{1,2\}, S_1, S_2, u_1, u_2 \rangle$ is a two person strategic form game. If the strategies of the players can be regulated by binding contracts, then one possibility for the feasible set F is the set of all allocations under correlated strategies:

$$F = \{(u_1(\alpha), u_2(\alpha)) : \alpha \in \Delta(S_1 \times S_2)\} \quad \text{where}$$

$$u_i(\alpha) = \sum_{s \in S} \alpha(s) u_i(s)$$

We could also choose the set of allocations under individually rational correlated strategies (recall that the payoffs for players under individually rational correlated strategies will be at least the respective minmax values). If the players' strategies cannot be regulated by binding contracts then a possibility for F would be the set of all allocations under correlated equilibria:

$$F = \{(u_1(\alpha), u_2(\alpha)) : \alpha \text{ is a correlated equilibrium of } \Gamma\}$$

There are several possibilities for the disagreement point v. The first possibility is to let v_i be the minmax value for player i.

$$v_1 = \min_{\sigma_2 \in \Delta(S_2)} \max_{\sigma_1 \in \Delta(S_1)} u_1(\sigma_1, \sigma_2)$$

$$v_2 = \min_{\sigma_1 \in \Delta(S_1)} \max_{\sigma_2 \in \Delta(S_2)} u_2(\sigma_1, \sigma_2)$$

The above choice is quite reasonable and scientific because the minmax value for a player is the minimum guaranteed payoff to the player even in a strictly competitive setting where the other player always tries to hurt this player the most.

A second possibility is to choose some focal Nash equilibrium (σ_1^*, σ_2^*) of Γ and let

$$v_1 = u_1(\sigma_1^*, \sigma_2^*); \quad v_2 = u_2(\sigma_1^*, \sigma_2^*)$$

A focal Nash equilibrium is one which becomes a natural choice for the players due to some external factors prevailing at the time of allocation.

A third possibility is to derive $v = (v_1, v_2)$ from some *rational threats*. The theory of rational threats has been proposed using rationality-based arguments and the reader is referred to the book by Myerson [3] for more details.

A sound theory of negotiation or arbitration must allow us to identify, given any bargaining problem (F, v), a unique allocation vector in $\mathbb{R}^2$ that would be selected as a result of negotiation or arbitration. Let us denote this unique allocation by $f(F, v)$. Thus the bargaining problem involves finding an appropriate solution function $f(.)$ from the set of all two-person bargaining problems into $\mathbb{R}^2$, which is the set of payoff allocations.

26.3 The Nash Axioms

John Nash used a brilliant axiomatic approach to solve this problem. He first came up with a list of properties an ideal bargaining solution function is expected to satisfy and then proved that there exists a unique solution that satisfies all of these properties. The following are the five axioms of Nash:

(1) Strong Efficiency
(2) Individual Rationality
(3) Scale Covariance
(4) Independence of Irrelevant Alternatives (IIA)
(5) Symmetry

Let us use the notation $f(F, v) = (f_1(F, v), f_2(F, v))$ to denote the Nash bargaining solution for the bargaining problem (F, v).

Axiom 1: Strong Efficiency

Given a feasible set F, we say an allocation $x = (x_1, x_2) \in F$ is *strongly Pareto efficient* or simply *strongly efficient* iff there exists no other point $y = (y_1, y_2) \in F$ such that $y_1 \geq x_1;\ \ y_2 \geq x_2$ with strict inequality satisfied for at least one player. An allocation $x = (x_1, x_2) \in F$ is *weakly Pareto efficient* or *weakly efficient* iff there exists no other point $y = (y_1, y_2) \in F$ such that $y_1 > x_1;\ \ y_2 > x_2$.

The strong efficiency axiom asserts that the solution to any two person bargaining problem should be feasible and strongly Pareto efficient. Formally, $f(F, v) \in F$ and there does not exist any $x = (x_1, x_2) \in F$ such that $x_1 \geq f_1(F, v);\ \ x_2 \geq f_2(F, v)$ with $x_i > f_i(F, v)$ for at least some $i \in \{1, 2\}$. This implies that there should be no other feasible allocation that is better than the solution for one player and not worse than the solution for the other player.

Axiom 2: Individual Rationality

This axiom states that $f(F, v) \geq v$ which implies that $f_1(F, v) \geq v_1;\ \ f_2(F, v) \geq v_2$. This means, neither player should get less in the bargaining solution than he/she could get in disagreement. Figure 26.1 illustrates Axioms 1 and 2.

Axiom 3: Scale Covariance

This axiom is stated as follows. For any numbers $\lambda_1, \lambda_2, \mu_1, \mu_2$ with $\lambda_1 > 0$, $\lambda_2 > 0$, define the set $G = \{(\lambda_1 x_1 + \mu_1, \lambda_2 x_2 + \mu_2 : (x_1, x_2) \in F\}$ and the point $w = (\lambda_1 v_1 + \mu_1, \lambda_2 v_2 + \mu_2)$. Then the solution $f(G, w)$ for the problem (G, w) is given by

$$f(G, w) = (\lambda_1 f_1(F, v) + \mu_1, \lambda_2 f_2(F, v) + \mu_2)$$

In the above, the bargaining problem (G, w) can be derived from (F, v) by applying an affine utility transformations which will not affect relevant properties of the utility

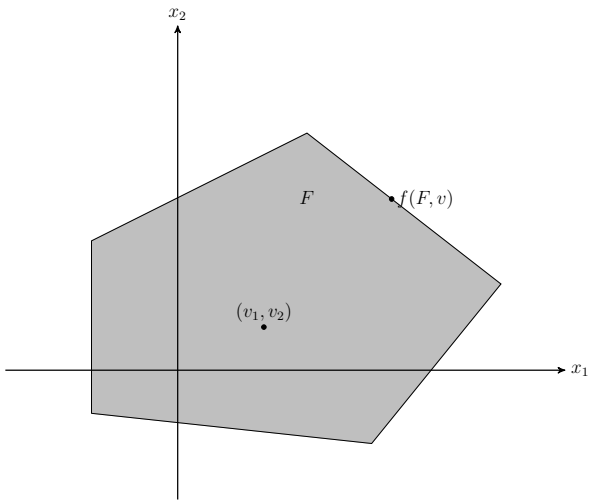

Fig. 26.1: Illustrating strong efficiency and individual rationality

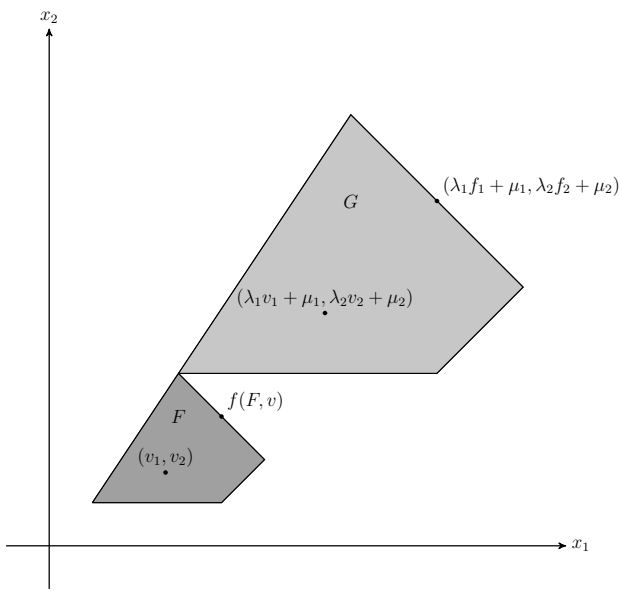

Fig. 26.2: Illustration of scale covariance

functions. The axiom implies that the solution of (G, w) can be derived from that of (F, v) by the same transformation. Figure 26.2 illustrates this axiom.

Axiom 4: Independence of Irrelevant Alternatives

This axiom states that, for any closed convex set G,

$$G \subseteq F \text{ and } f(F, v) \in G \Longrightarrow f(G, v) = f(F, v)$$

The axiom asserts that eliminating feasible alternatives (other than the disagreement point) that would not have been chosen should not affect the solution. The eliminated alternatives are referred to as irrelevant alternatives. This axiom is illustrated in Figure 26.3. If a mediator or referee were to select a solution by maximizing some aggregate measure of social gain, that is,

$$f(F, v) = \max_{x \in F} M(x, v)$$

where $M(x, v)$ is a measure of social gain by choosing x instead of v, then Axiom 4 can be shown to be always satisfied.

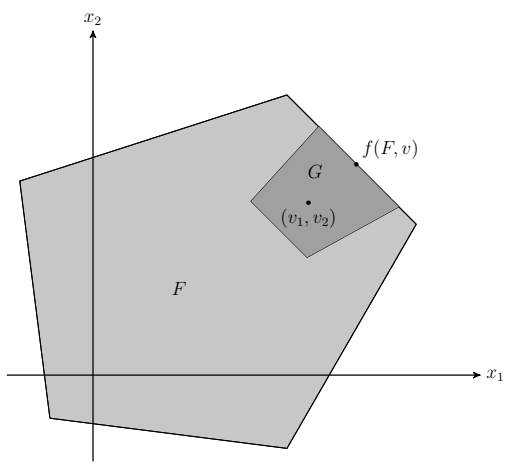

Fig. 26.3: Independence of irrelevant alternatives

Axiom 5: Symmetry

This axiom asserts that if the positions of players 1 and 2 are completely symmetric in the bargaining problem, then the solution should also treat them symmetrically. Formally,

$$v_1 = v_2 \text{ and } \{(x_2, x_1) : (x_1, x_2) \in F\} = F \Longrightarrow f_1(F, v) = f_2(F, v)$$

This axiom is illustrated in Figure 26.4.

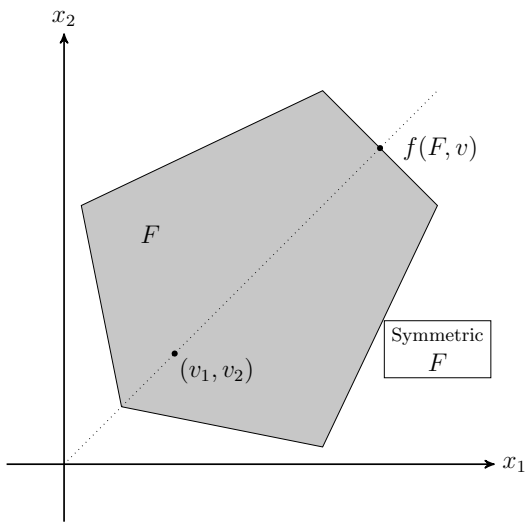

Fig. 26.4: Illustration of symmetry

26.4 The Nash Bargaining Solution

Using ingenious arguments, Nash showed that there exists a unique solution that satisfies all the five axioms. The following theorem presents the solution.

Theorem 26.1. *Given a two person bargaining problem (F, v), there exists a unique solution function $f(.,.)$ that satisfies Axioms (1) through (5). The solution function satisfies, for every two person bargaining problem (F, v),*

$$f(F, v) \in \underset{\substack{(x_1, x_2) \in F \\ x_1 \geq v_1; x_2 \geq v_2}}{\operatorname{argmax}} \big((x_1 - v_1)(x_2 - v_2)\big)$$

An Illustrative Example

Figure 26.5 shows a closed convex set F which is actually the convex hull enclosing the points $(4, 0)$, $(1, 1)$, and $(0, 4)$. Suppose F is the feasible set and $(1, 1)$ is the default point. It can be shown that the Nash bargaining solution is $(2, 2)$. This illustrates Pareto efficiency, individual rationality, and symmetry. We shall define a new feasible space G obtained by the following scaling: $\lambda_1 = \lambda_2 = \frac{1}{2}$; $\mu_1 = \mu_2 = 1$. Consider the bargaining problem (G, w) with $w = (\frac{3}{2}, \frac{3}{2})$ that is obtained using $w = (\lambda_1 v_1 + \mu_1, \ \lambda_2 v_2 + \mu_2)$. Using scale covariance, the Nash bargaining solution becomes $(2, 2)$. If H is the feasible space obtained using $\lambda_1 = \lambda_2 = \frac{1}{2}$; $\mu_1 = \mu_2 = 0$, then the problem $(H, (\frac{1}{2}, \frac{1}{2}))$ illustrates another instance of scale covariance.

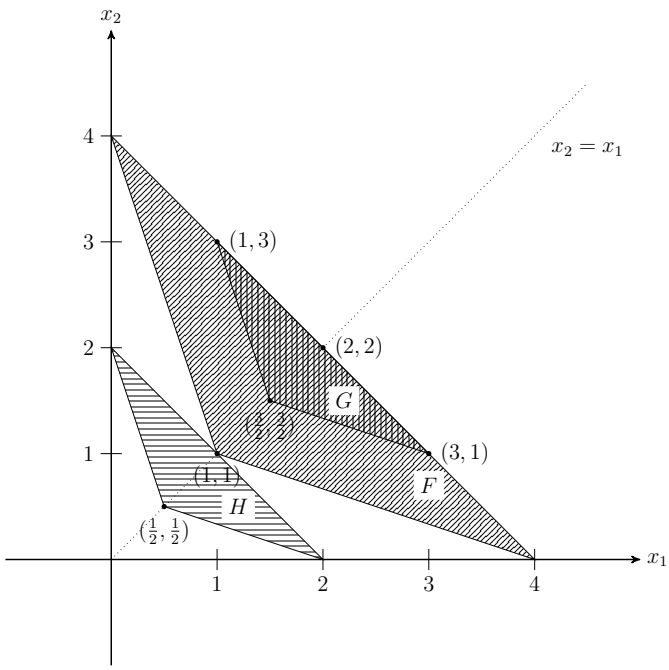

Fig. 26.5: An example to illustrate Nash axioms

26.5 Proof of the Nash Bargaining Theorem

First we prove the theorem for a special class of two person bargaining problems called *essential* bargaining problems and subsequently, we generalize this to the entire class of problems. A two person bargaining problem (F, v) is said to be *essential* if there exists at least one allocation $y \in F$ that is strictly better for both the players than the disagreement allocation v, that is, $y_1 > v_1$ and $y_2 > v_2$.

Proof for Essential Bargaining Problems

We are given an essential two person bargaining problem (F, v). Clearly, there exists some $y = (y_1, y_2) \in F$ such that $y_1 > v_1$ and $y_2 > v_2$.

Recall the definition of a quasiconcave function from Chapter 10: A function $f : S \to \mathbb{R}$ where S is non-empty and convex is said to be quasiconcave if

$$f(\lambda x + (1-\lambda)y) \geq \min(f(x), f(y)) \;\; \forall x, y \in S \;\; \forall \lambda \in [0,1]$$

f is strictly quasiconcave if

$$f(\lambda x + (1-\lambda)y) > \min(f(x), f(y)) \;\; \forall x, y \in S \;\; x \neq y \;\; \forall \lambda \in (0,1)$$

It is known that a strictly quasiconcave function will have a unique optimal solution

(maximum). Consider the optimization problem:

$$\begin{aligned} \max \quad & \big((x_1 - v_1)(x_2 - v_2)\big) \\ (x_1, x_2) \in F & \\ x_1 \geq v_1; x_2 \geq v_2 & \end{aligned}$$

where $(v_1, v_2) \in F$. The function $(x_1 - v_1)(x_2 - v_2)$ is strictly quasiconcave (since we are dealing with an essential bargaining problem) and therefore it has a unique maximizer. Therefore the above optimization problem has a unique optimal solution. Call this solution as (x_1^*, x_2^*).

Let $F \subset \mathbb{R}^2$ be convex and closed. Suppose $v = (v_1, v_2) \in F$ and

$$F \cap \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq v_1; x_2 \geq v_2\}$$

is non-empty and bounded. Suppose the function $f(F, v)$ satisfies the five axioms (1) strong efficiency (2) individual rationality (3) scale covariance (4) independence of irrelevant alternatives, and (5) symmetry. Then the Nash bargaining theorem states that $f(F, v) = (x_1^*, x_2^*)$ which happens to be the unique solution of optimization problem above.

The proof of the Nash bargaining theorem proceeds in two parts. In Part 1, we show that the optimal solution (x_1^*, x_2^*) of the optimization problem above satisfies all the five axioms. In Part 2, we show that if $f(F, v)$ satisfies all the five axioms, then $f(F, v) = (x_1^*, x_2^*)$. We use the following notation for the objective function in the rest of this proof:

$$N(x_1, x_2) = (x_1 - v_1)(x_2 - v_2)$$

The objective function is appropriately called the *Nash Product*.

Proof of Part 1

We have to show that the optimal solution (x_1^*, x_2^*) of the optimization problem above satisfies all the five axioms. We do this sequentially.

Strong Efficiency

We have to show that there does not exist $(\hat{x_1}, \hat{x_2}) \in F$ such that $\hat{x_1} \geq x_1^*$ and $\hat{x_2} \geq x_2^*$ with at least one inequality strict. Suppose such a $(\hat{x_1}, \hat{x_2})$ exists. Since there exists a $(y_1, y_2) \in F$ such that $y_1 > v_1$ and $y_2 > v_2$, the maximum value of the Nash product in the optimization problem is strictly greater than zero. Since the objective function $N(x_1, x_2)$ is increasing in x_1 and x_2, we have

$$N(\hat{x_1}, \hat{x_2}) > N(x_1^*, x_2^*)$$

which is not possible since $N(x_1^*, x_2^*)$ is the maximum possible value of $N(x_1, x_2)$ in the optimization problem.

Individual Rationality

This is immediately satisfied, being one of the constraints in the optimization problem.

Scale Covariance

For $\lambda_1 > 0, \lambda_2 > 0, \mu_1, \mu_2$, define

$$G = \{\lambda_1 x_1 + \mu_1, \lambda_2 x_2 + \mu_2) : (x_1, x_2) \in F\}$$

Consider the problem

$$\max_{(y_1,y_2)\in G} (y_1 - (\lambda_1 v_1 + \mu_1))(y_2 - (\lambda_2 v_2 + \mu_2))$$

This can be written using $y_1 = \lambda_1 x_1 + \mu_1$ and $y_2 = \lambda_2 x_2 + \mu_2$ as

$$\max_{(x_1,x_2)\in G} (\lambda_1 x_1 + \mu_1 - (\lambda_1 v_1 + \mu_1))(\lambda_2 x_2 + \mu_2 - (\lambda_2 v_2 + \mu_2))$$

The above problem is the same as

$$\max_{(x_1,x_2)\in G} \lambda_1 \lambda_2 (x_1 - v_1)(x_2 - v_2)$$

which attains maximum at (x_1^*, x_2^*). Therefore the problem

$$\max_{(y_1,y_2)\in G} (y_1 - (\lambda_1 v_1 + \mu_1))\ (y_2 - (\lambda_2 v_2 + \mu_2))$$

attains maximum at $(\lambda_1 x_1^* + \mu_1,\ \lambda_2 x_2^* + \mu_2)$.

Independence of Irrelevant Alternatives

We are given $G \subseteq F$ with G closed and convex. Let (x_1^*, x_2^*) be optimal to (F, v) and let (y_1^*, y_2^*) be optimal to (G, v). It is also given that $(x_1^*, x_2^*) \in G$.

- Since (x_1^*, x_2^*) is optimal to F which is a superset of G, we have
$$N(x_1^*, x_2^*) \geq N(y_1^*, y_2^*)$$
- Since (y_1^*, y_2^*) is optimal to G and $(x_1^*, x_2^*) \in G$, we have
$$N(y_1^*, y_2^*) \geq N(x_1^*, x_2^*)$$

Therefore we have

$$N(x_1^*, x_2^*) = N(y_1^*, y_2^*)$$

Since the optimal solution is unique, we then immediately obtain

$$(x_1^*, x_2^*) = (y_1^*, y_2^*)$$

Symmetry

Suppose we have $\{(x_2, x_1) : (x_1, x_2) \in F\} = F$ and $v_1 = v_2$. Since $v_1 = v_2$, we can use v_1 and v_2 interchangeably. We can therefore say that (x_1^*, x_2^*) maximizes $(x_1 - v_2)(x_2 - v_1)$. Since the optimal solution is unique, we should have $(x_1^*, x_2^*) = (x_2^*, x_1^*)$ which immediately yields $x_1^* = x_2^*$.

Proof of Part 2

We are given that $f(F, v)$ is a bargaining solution that satisfies all the five axioms. We have to show that $f(F, v)$ is the same as (x_1^*, x_2^*) which is the unique solution of the optimization problem:

$$\max_{(x_1, x_2) \in F, \; x_1 \geq v_1; x_2 \geq v_2} \big((x_1 - v_1)(x_2 - v_2)\big)$$

We have that $x_1^* > v_1$ and $x_2^* > v_2$. Consider the following transformation

$$L(x_1, x_2) = (\lambda_1 x_1 + \mu_1, \lambda_2 x_2 + \mu_2)$$

where

$$\lambda_1 = \frac{1}{x_1^* - v_1}; \;\; \lambda_2 = \frac{1}{x_2^* - v_2}; \;\; \mu_1 = \frac{-v_1}{x_1^* - v_1}; \;\; \mu_2 = \frac{-v_2}{x_2^* - v_2}$$

In other words, we have

$$L(x_1, x_2) = \Big(\frac{x_1 - v_1}{x_1^* - v_1}, \;\; \frac{x_2 - v_2}{x_2^* - v_2}\Big)$$

Note that $L(v_1, v_2) = (0, 0)$ and $L(x_1^*, x_2^*) = (1, 1)$. Defining

$$G = \{L(x_1, x_2) : (x_1, x_2) \in F\},$$

we have thus transformed the problem $(F, (v_1, v_2))$ to the problem $(G, (0, 0))$. In fact, since $L(x_1^*, x_2^*) = (1, 1)$, it is known that the transformed problem $(G, (0, 0))$ has its solution at $(1, 1)$. See Figure 26.6. Also observe that the objective function of the transformed problem is $(x_1 - 0)(x_2 - 0) = x_1 x_2$.

It can be shown that

$$x_1 + x_2 \leq 2 \;\; \forall (x_1, x_2) \in G$$

To show this, we assume that $x_1 + x_2 > 2$ and we can arrive at a contradiction. We leave this as an exercise.

G is bounded and we can always find a rectangle H which is symmetric about the line $x_1 = x_2$ such that $G \subseteq H$ and H is also convex and bounded. Further choose H such that the point $(1, 1) \in G$ is on the boundary of H.

Strong efficiency and symmetry imply that

$$f(H, (0, 0)) = (1, 1)$$

We can now use independence of irrelevant alternatives to get

$$f(G, (0, 0)) = (1, 1)$$

We know G is obtained through scaling. Now use scale covariance to get

$$f(G, (0, 0)) = L(f(F, v))$$

This implies

$$L(f(F, v)) = (1, 1)$$

Since we know that $L(x_1^*, x_2^*) = (1, 1)$, we finally obtain

$$f(F, v) = (x_1^*, \;\; x_2^*)$$

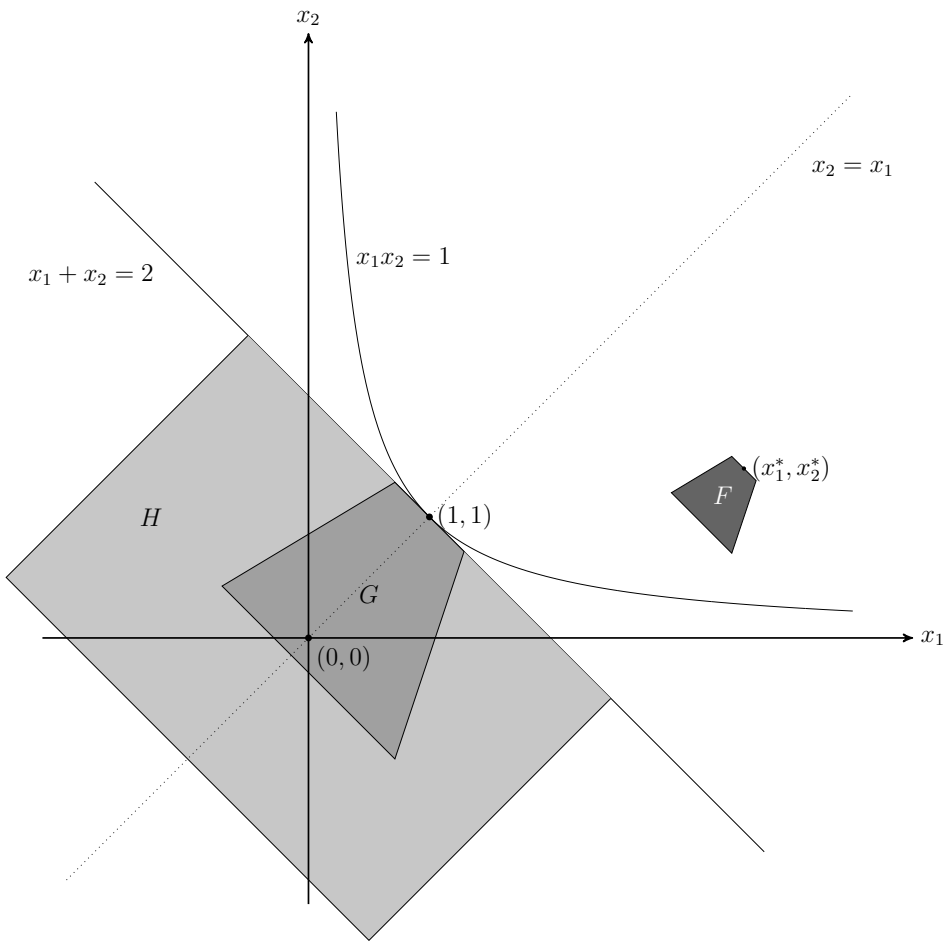

Fig. 26.6: Diagram showing F, G, H for proving the Nash bargaining theorem

Proof for Inessential Bargaining Problems

Consider the case when the problem (F, v) is inessential, that is, there does not exist any $(y_1, y_2) \in F$ such that $y_1 > v_1$; $y_2 > v_2$. Since F is convex, the above implies that there exists at least one player i such that

$$y_1 \geq v_1 \text{ and } y_2 \geq v_2 \Longrightarrow y_i = v_i \ \ \forall\, (y_1, y_2) \in F$$

The reason is if we could find $(y_1, y_2), (z_1, z_2) \in F$ such that $(y_1, y_2) \geq (z_1, z_2), y_1 > v_1, z_2 > v_2$, then $\frac{1}{2}(y_1, y_2) + \frac{1}{2}(z_1, z_2)$ would be a point in F that is strictly better than (v_1, v_2) for both the players. In the rest of the discussion, let us assume without loss of generality that

$$y_1 \geq v_1 \text{ and } y_2 \geq v_2 \Longrightarrow y_1 = v_1 \ \ \forall\, (y_1, y_2) \in F$$

Suppose x^* is an allocation in F that is best for the player 2 subject to the constraint $x_1 = v_1$ (with this constraint, the maximum value of the Nash product in the

optimization problem will be zero). This would imply that the vector (x_1^*, x_2^*) is the unique point that is strongly efficient in F and individually rational relative to v. This would mean that, to satisfy Axioms 1 and 2, we must have $f(F, v) = (x_1^*, x_2^*)$. It can be easily observed that (x_1^*, x_2^*) achieves the maximum value of the Nash product $(x_1 - v_1)(x_2 - v_2)$ which happens to be zero for all individually rational allocations. ■

Note. For essential bargaining problems, the strong efficiency axiom can be replaced by the following weak efficiency axiom:

Axiom 1′. (Weak Efficiency). $f(F, v) \in F$ and there does not exist any $y \in F$ such that

$$y > f(F, v).$$

Also, it is trivial to see that for essential bargaining problems, the individual rationality assumption is not required. Thus in the case of essential two person bargaining problems, any solution that satisfies Axioms $1', 3, 4, 5$ must coincide with the Nash bargaining solution.

26.6 Egalitarian and Utilitarian Solutions

We now discuss two other well known solution approaches to the bargaining problem, namely egalitarian and utilitarian solutions [3]. In typical bargaining situations, interpersonal comparisons of utility are made in two different ways.

- *Principle of Equal Gains*: Here a person's argument will be: "You should do this for me because I am doing more for you." This leads to what is called an egalitarian solution.
- *Principle of the Greatest Good*: The argument here goes as follows: "You should do this for me because it helps me more than it hurts you." This leads to the so called utilitarian solution.

For a two-person bargaining problem (F, v), the egalitarian solution is the unique point $(x_1, x_2) \in F$ that is weakly efficient in F and satisfies the equal gains condition:

$$x_1 - v_1 = x_2 - v_2$$

Recall that $(x_1, x_2) \in F$ is said to be weakly efficient if there does not exist any $(y_1, y_2) \in F$ such that $y_1 > x_1$ and $y_2 > x_2$.

A utilitarian solution is any solution function that selects, for every two person bargaining problem (F, v), an allocation $(x_1, x_2) \in F$ such that

$$x_1 + x_2 = \max_{(y_1, y_2) \in F} (y_1 + y_2)$$

If agents in negotiation or arbitration are guided by the equal gains principle, the natural outcome is the egalitarian solution. If it is guided by the principle of greatest good, a natural outcome is a utilitarian solution.

It can be shown that the egalitarian and the utilitarian solutions violate the axiom of scale covariance.

λ-Egalitarian Solution

Consider a two person bargaining problem (F, v). Given numbers $\lambda_1, \lambda_2, \mu_1, \mu_2$, with $\lambda_1 > 0$, $\lambda_2 > 0$, let

$$L(y) = (\lambda_1 y_1 + \mu_1, \lambda_2 y_2 + \mu_2) \text{ for } y \in \mathbb{R}^2$$

Given the problem (F, v), define

$$L(F) = \{L(y) : y \in F\}$$

Then the egalitarian solution of $(L(F), L(v))$ is $L(x)$, where x is the unique weakly efficient point in F such that

$$\lambda_1(x_1 - v_1) = \lambda_2(x_2 - v_2)$$

This is called the λ-egalitarian solution of (F, v). If $\lambda = (1, 1)$, this is called the simple egalitarian solution. The egalitarian solution does not satisfy scale covariance because the λ-egalitarian solution is generally different from the simple egalitarian solution.

λ-Utilitarian Solution

A utilitarian solution of $(L(F), L(v))$ is a point $L(z)$ where $z = (z_1, z_2)$ is a point in F such that

$$\lambda_1 z_1 + \lambda_2 z_2 = \max_{(y_1, y_2) \in F} (\lambda_1 y_1 + \lambda_2 y_2)$$

The solution point z is called a λ-utilitarian solution of (F, v). Utilitarian solutions do not satisfy scale covariance because a λ-utilitarian solution is generally not a simple utilitarian solution.

Relationship to the Nash Bargaining Solution

Note that the equal gains principle suggests a family of egalitarian solutions and the greatest good principle suggests a family of utilitarian solutions. These solutions correspond to application of these principles when the payoffs are compared in a λ-weighted utility scale.

As λ_1 increases and λ_2 decreases, the λ-egalitarian solutions will present us with an individually rational, weakly efficient frontier moving in the direction of decreasing payoff to player 1. Also as λ_1 increases and λ_2 decreases, the λ-utilitarian solutions will present us with an entire weakly efficient frontier, moving in the direction of increasing payoff to player 1.

It turns out that for an essential two person bargaining problem (F, v), there exists a vector $\lambda = (\lambda_1, \lambda_2)$ such that $\lambda > (0, 0)$ and the λ-egalitarian solution of

(F, v) is also a λ-utilitarian solution of (F, v). λ_1 and λ_2 that satisfy this property are called *natural scale factors*.

Remarkably, the allocation in F that is both λ-egalitarian and λ-utilitarian in terms of the natural scale factors is the Nash bargaining solution. Thus the Nash bargaining solution can be viewed as a natural synthesis of the equal gains and greatest good principles. The following proposition formalizes this fact.

Proposition 26.1. *Let (F, v) be an essential two person bargaining problem. Suppose x is an allocation vector such that $x^* \in F$ and $x^* \geq v$. Then x^* is the Nash-bargaining solution for (F, v) iff there exist strictly positive numbers λ_1 and λ_2 such that*

$$\lambda_1 x_1^* - \lambda_1 v_1 = \lambda_2 x_2^* - \lambda_2 v_2; \quad \text{and}$$

$$\lambda_1 x_1^* + \lambda_2 x_2^* = \max_{y \in F} (\lambda_1 y_1 + \lambda_2 y_2)$$

The reader is referred to the book by Myerson [3] for a proof of this result.

26.7 Summary and References

We summarize the main results of this chapter below.

- The two person bargaining problem $(F, (v_1, v_2))$ involves finding a satisfactory allocation of payoffs to two players, given a closed, convex set F of possible payoff allocations and a disagreement allocation (v_1, v_2) (that will be enforced in the event of disagreement). It is also assumed that unbounded gains over the disagreement allocation are not allowed.
- John Nash solved this problem using an axiomatic approach that mandates five axioms to be satisfied by any solution: (1) Pareto efficiency (2) individual rationality (3) scale covariance (4) independence of irrelevant alternatives (5) symmetry. Nash showed that there exists exactly one solution that satisfies all the five properties and it is the unique solution of the simple constrained optimization problem:

$$\begin{aligned} \max \quad & \big((x_1 - v_1)(x_2 - v_2)\big) \\ (x_1, x_2) \in F & \\ x_1 \geq v_1; x_2 \geq v_2 & \end{aligned}$$

- There are several other approaches for solving the two person bargaining problem. Two of these were discussed: (a) egalitarian approach and (b) utilitarian approach. The solutions produced by these two approaches would be the same and in fact coincide with the Nash bargaining solution, for a particular natural choice of scale factors.

The original discussion of the Nash bargaining problem and its solution is of course found in the classic paper of Nash [1]. The discussion and approach presented in this chapter closely follows that of Myerson [3]. There are two books that deal with the bargaining problem quite extensively. The book by Muthoo [4] deals exclusively with the bargaining problem. The book by Osborne and Rubinstein [5] discusses bargaining theory extensively and presents rigorous applications to market situations of different kinds. The book by Maschler, Solan, and Zamir [6] is a rich source of material for the Nash bargaining problem. The book by Straffin [7] provides two real-world applications of the two person bargaining problem: (1) management - labor union negotiations, and (2) duopoly Model of two competing companies trying to maximize their revenues. The concept of Nash program is discussed by Nash [2].

The Nash bargaining solution does not generalize to the case of three or more players. We bring out this fact in the next chapter and start studying other solution concepts in cooperative game theory.

References

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26.8 Exercises

(1) With reference to the axiom on independence of irrelevant alternatives, show the following property: Given a Nash bargaining problem (F, v), if an arbitrator were to select a solution by maximizing some aggregate measure of social gain, that is,

$$f(F, v) = \max_{x \in F} M(x, v)$$

where $M(x, v)$ is a measure of social gain by choosing x instead of v, then Axiom 4 can be shown to be always satisfied.

(2) Suppose F is the convex hull enclosing the points $A = (1, 8)$; $B = (6, 7)$; $C = (8, 6)$; $D = (9, 5)$; $E = (10, 3)$; $F = (11, -1)$; and $G = (-1, -1)$. Suppose the default point (status quo point) is $(2, 1)$. Compute the Nash bargaining solution for this situation. Draw a picture and that will be helpful.

(3) In the above problem, how will the Nash bargaining solution change when you choose each of the points A, B, C, D, E, F, G as the default point?

(4) Provide an example of a two player bargaining problem which has the same Nash bargaining solution irrespective of the default point chosen.

(5) In the proof of Part 2 of the Nash bargaining theorem, show the following result:

$$x_1 + x_2 \leq 2 \;\; \forall (x_1, x_2) \in G$$

Hint: Assume that $x_1 + x_2 > 2$ and arrive at a contradiction.

(6) Consider the following two player game:

	A	B
A	$5, 5$	$2, 2$
B	$2, 2$	$3, 3$

For this game,

- Compute the set of all payoff utility pairs possible (a) under correlated strategies (b) under individually rational correlated strategies.
- What would be the Nash bargaining solution in cases (a) and (b) assuming the minmax values as the disagreement points?